Polynomial-Linked Tucker Models for Nonlinear Low-Rank Tensor Observations

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Abstract

Classical Tucker models explain multiway data with a multilinear low-rank structure. This is effective when the observed tensor is close to a multilinear signal, but many sensing and imaging pipelines include nonlinear entry-level effects, such as response curves, saturation, and material-dependent distortions. These effects can make a low-rank latent tensor appear higher-rank in the observed domain.

We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL), a simple extension of Tucker for such observations. NTD-PL first forms a shared Tucker latent tensor and then maps each latent entry through a finite polynomial link. The model contains standard Tucker as the degree-one case and adds only a small number of scalar link coefficients. At the same time, powers of the latent tensor induce tied high-order interactions among the same Tucker factors, giving a compact rank-lift without introducing a full high-order tensor parameter.

We study the representation power of this model through approximation and separation results, and derive a masked learning objective for reconstruction and completion. The resulting solver alternates between Tucker backbone updates and a small ridge-regression step for the link coefficients, with degree continuation for stability. Experiments on controlled tensors, CAVE hyperspectral completion and reconstruction, and external HSI scenes show that NTD-PL improves matched-rank Tucker when the residual has a coherent nonlinear component, while also revealing boundary cases where the Tucker backbone already explains most of the signal.

1 Introduction

Tensor decompositions provide compact models for multiway data in scientific sensing, imaging, and signal processing [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013]. Tucker decomposition is a standard choice because it separates a small core from mode-wise factors, but its observation model is linear in the reconstructed latent entry. This can be too restrictive when a low-rank latent signal passes through saturation, response curves, calibration effects, or material-dependent nonlinearities before it is measured [Debevec and Malik, 1997, Grossberg and Nayar, 2004, Bioucas-Dias et al., 2012, Heylen et al., 2014]. After such an entry-level response, a simple latent Tucker tensor may appear higher-rank in the observed domain.

A direct remedy is to increase Tucker rank, but this expands the core and factors throughout the model. More flexible nonlinear tensor decoders based on kernels, Gaussian processes, or neural networks can also help, but they often make the relation to the Tucker backbone less transparent [Xu et al., 2011, Zhe et al., 2016, Liu et al., 2017, Saragadam et al., 2024]. We ask a narrower question: can a small observation link, learned jointly with the Tucker factors, make a tight multilinear rank budget more effective when the dominant residual is shared and entrywise?

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

We instantiate this idea with a polynomial link, a classical low-dimensional tool for modeling nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Let

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$$

be a Tucker latent tensor. Instead of fitting the observation directly by $\mathcal{S}$, we model

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p},$$

where powers are entrywise. We call the model Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case and adds only $P + 1$ scalar coefficients, while the powers $\mathcal{S}^{\odot p}$ induce tied high-order interactions among the same core and factors. Thus the link is not an independent high-rank correction; it is a compact way to expose nonlinear structure already organized by the Tucker backbone.

Our claims are intentionally scoped. NTD-PL is not meant to replace rank selection or general nonlinear tensor decoders. It targets settings where a shared entry-level response explains coherent residual structure left by a low-rank Tucker model.

Contributions. We make the following contributions.

- We propose NTD-PL, a Tucker model with a jointly learned polynomial observation link. Standard Tucker is recovered as the degree-one case.
- We characterize the model class through approximation results and a separation from standard Tucker at the same backbone rank.
- We derive a masked learning objective for reconstruction and completion, and use an alternating solver with a closed-form ridge update for the link coefficients.
- We evaluate the mechanism on controlled tensors and hyperspectral data, including completion, reconstruction, rank-lift, and post-hoc calibration checks.

2 Related Work

Low-rank tensor models. CP, Tucker, tensor train, and tensor ring decompositions provide compact representations for multiway data [Kolda and Bader, 2009, De Lathauwer et al., 2000, Oseledets, 2011, Zhao et al., 2016]. Tucker is the closest baseline for NTD-PL because its core and mode factors give the same multilinear backbone. Rather than changing ranks, factor structures, or regularization, we keep this backbone fixed and change the observation map from the latent tensor to the measured entries.

Nonlinear tensor factorization. Nonlinear tensor methods use generalized likelihoods, kernels, Gaussian processes, neural decoders, or nonlinear tensor networks [Hong et al., 2020, Xu et al., 2011, Zhe et al., 2016, Pan et al., 2020, Saragadam et al., 2024, Varolgunes et al., 2023]. These models can be more expressive than a scalar polynomial link, but they often trade away the simple finite-dimensional relation to Tucker. NTD-PL is complementary: it keeps the nonlinearity shared and entrywise so that its effect can be analyzed as a tied rank lift of the same Tucker factors.

Polynomial feature lifts and high-order interactions. Polynomial and Volterra-style expansions are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Related matrix and tensor methods use polynomial feature maps or symmetric high-order tensors to increase expressivity [Zhai et al., 2024, Liu et al., 2015, Deng et al., 2016]. Such lifts can expose nonlinear structure, but explicit high-order parameters often increase dimension and contraction cost. NTD-PL can be viewed as a shared-backbone compression of these interactions: the lifted features are induced by powers of one learned Tucker latent tensor, rather than by an independent high-order tensor parameter.

3 Model

3.1 Model and Observation Assumption

Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ be an observed tensor and let

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!], \qquad \mathcal{X} \in \mathbb{R}^{R_1 \times \cdots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \tag{1}$$

NTD-PL treats $\mathcal{S}$ as a shared low-rank latent tensor and models each observed entry through a scalar polynomial response:

$$Y_{\mathbf{i}} = f_{\boldsymbol{\beta}}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \qquad f_{\boldsymbol{\beta}}(s) = \sum_{p=0}^{P} \beta_p s^p, \qquad \mathbf{i} = (i_1, \ldots, i_N). \tag{2}$$

Equivalently, the tensor prediction is

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{P+1}. \tag{3}$$

The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel, and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the entry-generation map while leaving the Tucker backbone and its multilinear ranks fixed.

3.2 Shared-Backbone High-Order Interactions

Although $f_{\boldsymbol{\beta}}$ is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. We arrive at it by first defining an explicit p-fold Tucker extension: instead of mapping one latent index per mode, the mode-n projection is a higher-order tensor $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$ that couples p latent indices. The case $p = 1$ is ordinary Tucker, while $p = 2$ gives a quadratic Tucker-style interaction between two latent Tucker tuples. Appendix A.1 gives the formal entrywise construction and tensor-network view.

The explicit construction is a reference model, not the parameterization we train. Its projection tensors grow as $I_n R_n^p$, and each degree would add a new set of mode projections. NTD-PL keeps the same high-order interaction pattern by tying the p-fold projection to the original Tucker factor:

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)}, \ldots, r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}.$$

Under this tying, the degree-p channel is exactly $S_{\mathbf{i}}^p$. Thus it still couples p latent Tucker index tuples, but all copies share the same mode factors. Across degrees, the only new free parameters are the scalar coefficients $\{\beta_p\}$, which mix these tied channels.

The latent tensor $\mathcal{S}$ and the coefficients $\boldsymbol{\beta}$ are fitted jointly, so the low-rank backbone is optimized together with the polynomial channels. Section 4 analyzes the induced rank expansion and the resulting separation from standard Tucker.

3.3 Masked Least-Squares Objective

Let $\Omega \in \{0,1\}^{I_1 \times \cdots \times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_{\mathbf{i}} \Omega_{\mathbf{i}}$. We estimate the core, factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \boldsymbol{\beta}} \frac{1}{2|\Omega|} \left\| \Omega \odot (\mathcal{Y} - f_{\boldsymbol{\beta}}(\mathcal{S})) \right\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^{N} \frac{\lambda_n}{2} \left\| \mathbf{A}^{(n)} \right\|_F^2 + \frac{\lambda_\beta}{2} \|\boldsymbol{\beta}\|_2^2. \tag{4}$$

The same formulation covers full reconstruction by taking $\Omega \equiv \mathbf{1}$ and tensor completion by restricting the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the low-dimensional polynomial link.

4 Theory

The theory addresses two questions raised by the model definition. First, when the observation is generated by a shared entrywise nonlinearity, can a polynomial link approximate it without increasing the Tucker backbone rank? Second, what multilinear-rank structure is induced by the tied powers $\mathcal{S}^{\odot p}$? We answer these as expressivity questions about the model class.

4.1 Approximation Under Nonlinear Observations

Theorem 1 (Polynomial generation)**.** *Suppose $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1, \dots, R_N)$ and g is a polynomial of degree at most P. Then degree-P NTD-PL with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 2 (Analytic link approximation)**.** *Suppose $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1, \dots, R_N)$ and $\|\mathcal{S}^\star\|_\infty \le B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree-P NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \left\| \mathcal{Y} - \widehat{\mathcal{Y}} \right\|_F \le M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \qquad M = \prod_{n=1}^{N} I_n. \tag{5}$$

Thus a finite polynomial link gives a direct approximation route for smooth entrywise responses on a bounded latent range. The statement uses the simple Taylor truncation bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

4.2 Rank Expansion and Separation

Proposition 1 (Shared-backbone rank expansion)**.** *Let $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]\!]$ have Tucker rank at most $(R_1, \dots, R_N)$. For every integer $p \ge 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \tag{6}$$

More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original factors. For $\mathbf{A} \in \mathbb{R}^{I \times R}$, define $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{I \times D}$, $D = \binom{R+p-1}{p}$, by

$$A^{\langle p \rangle}_{i,\alpha} = \prod_{r=1}^{R} A_{i,r}^{\alpha_r}, \tag{7}$$

where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle}$.

This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree-p channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same backbone factors. The link therefore changes rank structure through tied features, not through an independent high-order parameter block.

Theorem 3 (Tensorial separation from standard Tucker)**.** *Let $N \ge 3$, $R \ge 2$, $p \ge 2$, and suppose $I_n \ge D$ for every n, where $D = \binom{R+p-1}{p}$. Then there exists an order-N tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ that is representable by a degree-p NTD-PL model with Tucker backbone rank $(R, \dots, R)$, but whose exact standard Tucker rank is*

$$(D, \dots, D). \tag{8}$$

Equivalently, a degree-p polynomial link can generate tensors whose multilinear ranks are $(\binom{R+p-1}{p}, \dots, \binom{R+p-1}{p})$ from a shared Tucker backbone of rank $(R, \dots, R)$.

The proof in Appendix B uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker tensors: it identifies a structured high-rank subset generated by a low-rank shared backbone.

Together, the results clarify the scope of NTD-PL. Polynomial links can approximate shared nonlinear observation maps while keeping the same latent Tucker rank, and their powers induce Veronese-lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without introducing untied high-order tensor parameters.

5 Optimization

We optimize the masked objective in Section 3.3 with a simple alternating scheme. The Tucker backbone $(\mathcal{X}, \{\mathbf{A}^{(n)}\})$ is updated by first-order steps through the current polynomial link, while the link coefficients are updated by a small ridge-regression problem on the current latent entries. This gives a practical solver while keeping the optimization layer standard.

Given the current latent tensor $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$, the coefficient update for active degree Q solves

$$\min_{\boldsymbol{\beta}_{0:Q}} \frac{1}{2|\Omega|} \left\| \mathbf{y}_\Omega - \mathbf{\Phi}_{\Omega,Q}(\mathcal{S}) \boldsymbol{\beta}_{0:Q} \right\|_2^2 + \frac{\lambda_\beta}{2} \left\| \boldsymbol{\beta}_{0:Q} \right\|_2^2, \tag{9}$$

where $\mathbf{\Phi}_{\Omega,Q}(\mathcal{S})$ contains the observed powers $[1, s, \ldots, s^Q]$. The backbone update uses the chain rule through $f_{\boldsymbol{\beta}}$; the implementation evaluates $f_{\boldsymbol{\beta}}(\mathcal{S})$ and $f'_{\boldsymbol{\beta}}(\mathcal{S})$ together, then backpropagates the masked residual through the Tucker reconstruction.

Training starts from the Tucker-compatible degree-one model and activates higher powers gradually. When a new degree is activated, its coefficient is initialized at zero, so the model expands from the current low-degree fit rather than beginning with all nonlinear terms active. We also normalize factor scales and absorb them into the core, which fixes a Tucker gauge without changing $\mathcal{S}$.

The dominant cost is the same forward and reverse Tucker contractions used by standard Tucker optimization at the chosen rank. The polynomial link adds $O(P|\Omega|)$ elementwise work and a $(P+1) \times (P+1)$ ridge solve, so for the small degrees used here its overhead is minor. Full update equations, stabilization details, and pseudocode are given in Appendix A.2.

6 Experiments

The empirical study tests a scoped mechanism: when a Tucker backbone is kept low-rank, can a shared polynomial link capture residual structure that is invisible to a linear Tucker observation model? The experiments are organized to separate this question from a broad benchmark claim. Controlled tensors isolate when the link should help, masked CAVE completion tests whether the learned low-rank geometry generalizes to held-out entries, CAVE reconstruction ablations rule out simple rank-lift and post-hoc calibration explanations, and external HSI scenes probe transfer and boundary cases.

Protocol. Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. NTD-PL is compared with Tucker at the same multilinear rank, except in the explicit rank-lift ablation where Tucker is given extra rank. We report RMSE and spectral angle (SAM), with NMSE(dB) used for cross-domain reconstruction where scale varies across tensors. Completion masks entries uniformly at random and computes all completion metrics only on the held-out entries.

6.1 Controlled Nonlinear Tensors

The first check isolates the intended effect. We generate tensors of the form $Y_\alpha = \sqrt{1-\alpha}\, S^\star + \sqrt{\alpha}\, \widetilde{R}^\star$, where $S^\star$ is low-rank Tucker and $\widetilde{R}^\star$ is the normalized component of a candidate nonlinear response that is orthogonal to $S^\star$. Thus α controls nonlinear residual energy rather than the amplitude of a raw scalar nonlinearity.

When the source is linear, NTD-PL reduces to Tucker up to the fitted affine link; when nonlinear residual energy is present, the higher-order link terms should become useful. Figure 1 shows this pattern across four residual families: NTD-PL separates from linear tensor baselines as α increases. Figure 2 then varies the maximum polynomial degree at fixed residual energy, showing that the gain is tied to the expressive order of the link. Degree-continuation curves in Appendix C.1 show that activating higher powers does not destabilize the fit. The formal residual construction and fitted link-term contributions are reported in Appendix C.1.

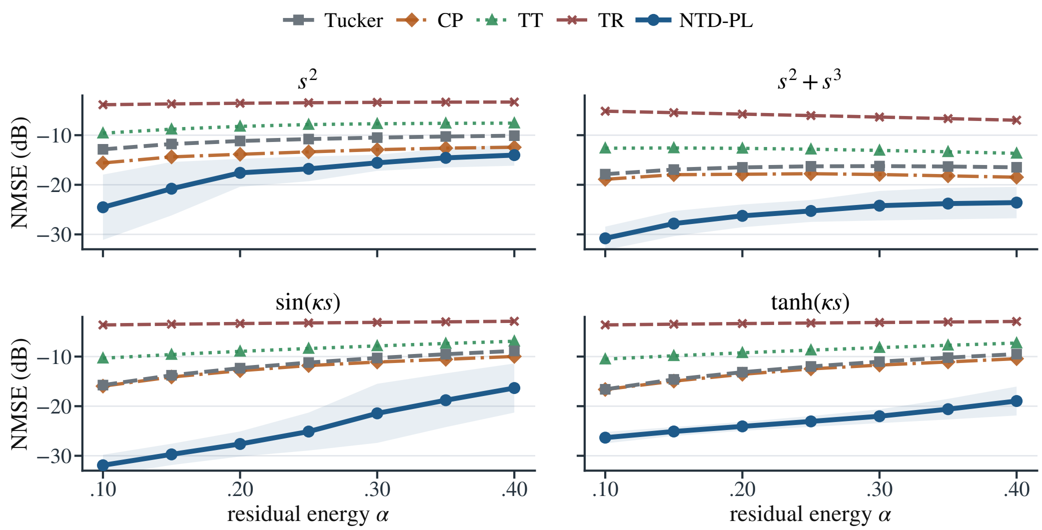

Figure 1: Controlled nonlinear alpha sweep. Each panel uses a different scalar response to define the orthogonal residual; lower NMSE(dB) is better.

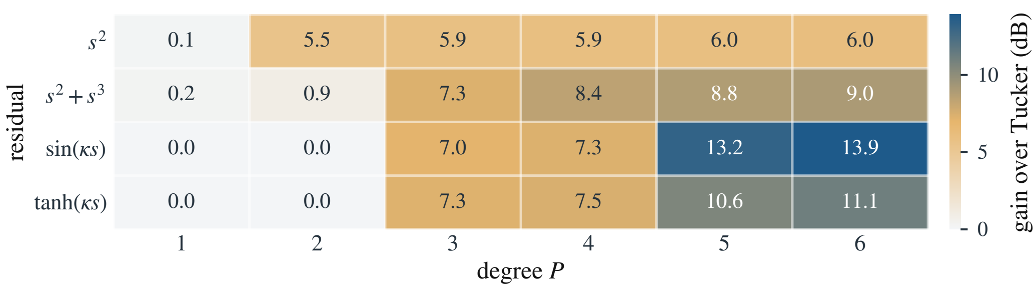

Figure 2: Controlled nonlinear degree-gain heatmap at $\alpha = 0.25$. Cell values are Tucker minus NTD-PL NMSE(dB), so larger values favor NTD-PL.

6.2 Low-Rank Masked Completion

Masked completion is the main empirical test because held-out entries expose whether the learned low-rank geometry transfers beyond the observed tensor. In this setting, the polynomial link is useful only if it changes the inductive bias of the shared Tucker backbone, rather than simply calibrating training entries. Table 1 shows that this effect persists across random-mask rates at full resolution: with the same rank budget as Tucker, NTD-PL gives consistent held-out RMSE gains and also improves spectral angle on most scenes. The scene-level sign and Wilcoxon tests in Appendix C.2 support the same conclusion.

Table 1: Full-resolution CAVE completion at rank $(33, 33, 4)$ under random masks. Metrics are computed on missing entries. Tucker and NTD-PL columns report mean $\pm$ standard deviation over scenes; gains are paired relative improvements over Tucker.

ρ	RMSE*↓			SAM*↓		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263 ± 0.0179	$\mathbf{0.0229\pm0.0158}$	12.9%	11.82 ± 5.33	$\mathbf{8.36\pm3.38}$	29.3%
0.3	0.0263 ± 0.0179	$\mathbf{0.0230\pm0.0158}$	12.5%	14.75 ± 6.33	$\mathbf{12.47\pm5.06}$	15.5%
0.5	0.0264 ± 0.0179	$\mathbf{0.0230\pm0.0158}$	12.9%	15.56 ± 6.85	$\mathbf{13.48\pm5.75}$	13.4%
0.7	0.0266 ± 0.0180	$\mathbf{0.0219\pm0.0128}$	17.7%	16.02 ± 7.21	$\mathbf{13.79\pm5.43}$	13.9%

6.3 CAVE Reconstruction Mechanism Checks

Full-observation reconstruction is a controlled mechanism check rather than the main generalization test: with all entries observed, we can ask whether the polynomial link consistently removes residual structure left by a matched-rank Tucker backbone. Table 2 supports this interpretation. NTD-PL improves both intensity error and spectral angle across the rank range, while the advantage becomes smaller as the Tucker rank increases. This trend is the expected signature of a lightweight observation model: it helps most when the linear backbone is tight, and becomes less necessary as ordinary multilinear capacity grows.

Table 2: CAVE full reconstruction at matched ranks. Metrics are averaged over 15 scenes.

Rank	Params	RMSE↓		SAM↓		Gain	
		Tucker	NTD-PL	Tucker	NTD-PL	RMSE	SAM
$(18, 18, 3)$	20k	0.0369	**0.0319**	22.36	**19.23**	13.6%	14.0%
$(24, 24, 4)$	27k	0.0299	**0.0256**	17.42	**15.15**	14.3%	13.0%
$(33, 33, 4)$	38k	0.0261	**0.0223**	16.43	**14.41**	14.3%	12.3%
$(40, 40, 5)$	49k	0.0217	**0.0192**	12.65	**11.85**	11.5%	6.4%

Two ablations clarify what the gain means. First, Table 3 asks how much spatial Tucker rank is needed before ordinary Tucker first matches the RMSE of a lower-rank NTD-PL model, while holding the spectral rank fixed. Tucker can buy back part of the RMSE gap by increasing rank, so NTD-PL should not be read as a replacement for rank selection. The spectral-angle gap, however, remains visible at the first RMSE-matching Tucker ranks, suggesting that the link is not merely trading for a slightly larger multilinear model. Second, post-hoc scalar calibration of a fitted Tucker reconstruction improves Tucker but remains below joint fitting in RMSE and SAM; the comparison and implementation details are given in Appendix C.3.

Table 3: Rank-lifted Tucker comparison on CAVE reconstruction. The table reports the first fixed-channel Tucker rank that matches each NTD-PL RMSE.

NTD-PL reference				First fixed-channel Tucker matching RMSE				Extra
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
$(18, 18, 3)$	20k	0.0319	**19.23**	$(28, 28, 3)$	31k	**0.0316**	21.41	$+59.5\%$
$(24, 24, 4)$	27k	0.0256	**15.15**	$(35, 35, 4)$	41k	**0.0254**	16.26	$+51.3\%$
$(33, 33, 4)$	38k	0.0223	**14.41**	$(49, 49, 4)$	60k	**0.0222**	15.28	$+56.5\%$

6.4 External HSI Validity and Boundary Cases

Table 4: External HSI reconstruction and completion. Columns report RMSE; gains are relative RMSE reductions over Tucker, and positive ΔSAM favors NTD-PL.

Dataset	Reconstruction				Completion			
	Tucker	NTD-PL	Gain	ΔSAM	Tucker	NTD-PL	Gain	ΔSAM
Jasper Ridge	0.0462	**0.0430**	**6.93%**	**0.98**	0.0471	**0.0434**	**7.82%**	**1.16**
Samson	0.0263	**0.0242**	**7.95%**	**0.06**	0.0268	**0.0244**	**8.86%**	**0.31**
Urban	0.0444	**0.0429**	**3.59%**	**0.29**	0.0454	**0.0433**	**4.67%**	**0.49**
Cuprite	**0.0176**	0.0177	-0.60%	-0.04	0.0179	**0.0178**	**0.56%**	-0.00

Table 4 asks whether the CAVE mechanism transfers to other hyperspectral tensors and where the shared-link assumption weakens. NTD-PL improves reconstruction and completion on Jasper Ridge, Samson, and Urban. Cuprite is the boundary case: Tucker is already strong, leaving little coherent residual for a shared polynomial link. Thus the evidence supports a scoped claim: NTD-PL is most useful under a tight rank budget when the linear Tucker residual is coherent enough to be modeled by a shared nonlinear link. Additional cross-domain checks are in Appendix C.5.

7 Limitations

NTD-PL assumes that the residual left by a low-rank Tucker backbone is well described by a shared entry-level nonlinear response. It is therefore not a universal tensor model: when Tucker already fits well at the chosen rank, or when residual nonlinearities vary strongly across space, spectra, or modes, a single scalar link may add little.

The method also does not remove rank selection. Increasing Tucker rank can recover part of the RMSE gap, and NTD-PL inherits Tucker's nonconvexity and initialization sensitivity. Although the link adds few parameters, evaluating it over all observed entries still increases runtime relative to plain Tucker.

8 Conclusion

We introduced NTD-PL, a Tucker model with a shared polynomial observation link. Rather than replacing multilinear low-rank structure, NTD-PL adds a small nonlinear layer that expresses structured residuals left by a linear Tucker fit. The theory explains this as tied high-order Tucker interactions, or a structured rank lift without an independent high-rank correction.

The experiments support the same scoped interpretation. Controlled tensors, held-out CAVE completion, reconstruction mechanism checks, and external HSI scenes show that the link helps when a shared nonlinear residual is plausible under a tight rank budget, while also revealing near-linear boundary cases. Adaptive links and broader tensor modalities are natural next steps.

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A Additional Model Details

A.1 Explicit p-Fold Tucker Motivation

This section expands the motivation behind the explicit p-fold Tucker channel in Section 3.2. The construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map. In the matrix case, one may move from a linear model $\mathbf{y} = \mathbf{A}\mathbf{x}$ to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

where the second term couples pairs of latent coordinates. The corresponding question for Tucker is: if standard Tucker is the multilinear low-rank map for multiway data, what is the direct p-th order interaction analogue?

For a Tucker backbone, an explicit p-fold interaction can be written as

$$\widetilde{Y}_{\mathbf{i}}^{(p)} = \sum_{\mathbf{r}^{(1)},\dots,\mathbf{r}^{(p)}} \left(\prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}} \right) \prod_{n=1}^{N} \mathcal{A}^{(n,p)}_{i_n, r_n^{(1)},\dots,r_n^{(p)}}. \tag{10}$$

Here each mode projection depends on p latent indices instead of one. For $p = 1$, this reduces to the ordinary Tucker entry formula. For $p = 2$, the same core appears twice and the mode projections couple two latent tuples, giving a quadratic Tucker-style channel. Figure 3 illustrates this progression.

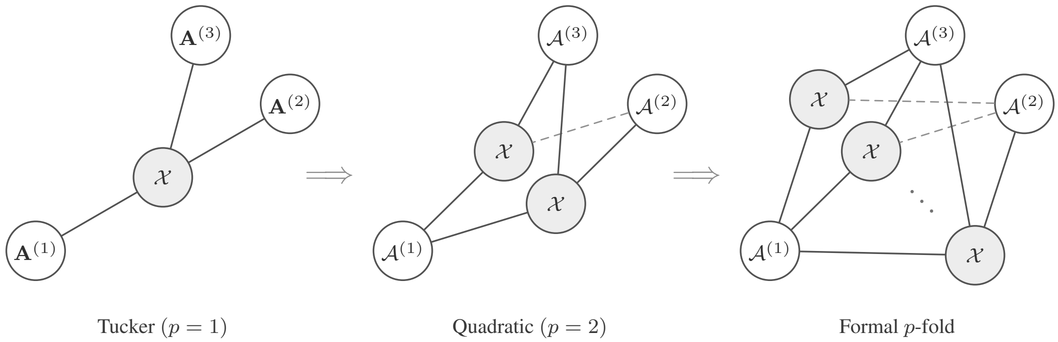

Figure 3: Tensor-network view of a formal p-fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

The explicit construction is useful as a reference model, but it is not the parameterization used by NTD-PL. Its mode projections $\mathcal{A}^{(n,p)}$ have size $I_n R_n^p$, so a direct implementation introduces new high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}. \tag{11}$$

Substituting (11) into (10) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]\!]$. Thus the degree-$p$ NTD-PL channel is a shared-projection version of the explicit p-fold Tucker prototype.

This is the modeling role of the p-fold Tucker prototype: it defines the high-order Tucker interaction that NTD-PL compresses. The final model does not train the untied tensors $\mathcal{A}^{(n,p)}$; it keeps a single Tucker backbone and adds only the polynomial coefficients $\{\beta_p\}$.

A.2 Optimization Details

This section gives the update equations behind the solver summarized in Section 5. The implementation follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression updates of the link coefficients.

Backbone gradients. For a fixed $\boldsymbol{\beta}$, define

$$f'_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=1}^{P} p \beta_p \mathcal{S}^{\odot (p-1)}. \tag{12}$$

The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_{\boldsymbol{\beta}}(\mathcal{S}) - \mathcal{Y}) \odot f'_{\boldsymbol{\beta}}(\mathcal{S}). \tag{13}$$

Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \cdots \times_N \mathbf{A}^{(N)\top} + \lambda_X \mathcal{X}. \tag{14}$$

For the n-th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \cdots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \cdots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \tag{15}$$

Then

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \tag{16}$$

The experiments use Adam for these first-order steps. This choice is not part of the model definition; any comparable first-order optimizer can use the same gradients.

Link update. With $\mathcal{S}$ fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is the ridge regression in (9). Its closed-form solution is

$$\boldsymbol{\beta}^{\star}_{0:Q} = \left(\boldsymbol{\Phi}_{\Omega,Q}^{\top}\boldsymbol{\Phi}_{\Omega,Q} + |\Omega|\lambda_{\beta}\mathbf{I}\right)^{-1}\boldsymbol{\Phi}_{\Omega,Q}^{\top}\mathbf{y}_{\Omega}. \tag{17}$$

Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In the code, this small system is solved either from an explicit Vandermonde design or from moment equations built from cached powers.

Stabilization and continuation. High powers of $\mathcal{S}$ can be sensitive to scale. The implementation therefore normalizes factor columns after backbone updates and absorbs the corresponding scales into the core; this is a gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For the coefficient solve, it may regress on the centered and scaled latent variable $(S-\mu)/\sigma$, then convert the solution back to the original power basis of S.

The active degree follows the schedule

$$t_k = \text{round}\left(\frac{kT}{P}\right), \qquad k = 1, \ldots, P-1.$$

When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new coefficient β_{p+1} is initialized at zero.

Cost. The dominant cost is forming the Tucker reconstruction and its reverse contractions on the observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same rank, plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms a Vandermonde design or the corresponding moment equations and solves a $(P+1)\times(P+1)$ linear system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the Tucker contractions.

Algorithm 1 gives the full procedural summary.

Algorithm 1 Masked NTD-PL optimization

Require: observed tensor $\mathcal{Y}$, mask Ω, rank $(R_1, \ldots, R_N)$, degree P

```
initialize 𝒳, {A^(n)} from Tucker on the masked tensor
initialize active degree P_act ← 1 and β = (0, 1)
for t = 1, ..., T do
    form 𝒮 = ⟦𝒳; A^(1), ..., A^(N)⟧
    if a link update is scheduled then
        update active coefficients β_{0:P_act} by ridge regression on {(S_i, Y_i) : Ω_i = 1}
    end if
    compute f_β(𝒮) and f'_β(𝒮)
    take first-order steps on 𝒳 and {A^(n)} using the masked gradient
    normalize factor scales and absorb them into the core
    increase P_act according to the continuation schedule, initializing the new coefficient at zero
end for
return 𝒳, {A^(n)}, β
```

B Additional Proofs

B.1 Polynomial and Analytic Approximation

Proof of Theorem 1. Since g has degree at most P, write

$$g(s) = \sum_{p=0}^{P} c_p s^p. \tag{18}$$

Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^\star$, and set $\beta_p = c_p$. Then, entrywise,

$$\widehat{\mathcal{Y}}_{i_1,\dots,i_N} = \sum_{p=0}^{P} \beta_p \left(\mathcal{S}^\star_{i_1,\dots,i_N}\right)^p = g\left(\mathcal{S}^\star_{i_1,\dots,i_N}\right) = \mathcal{Y}_{i_1,\dots,i_N}. \tag{19}$$

Thus $\widehat{\mathcal{Y}} = \mathcal{Y}$. $\square$

Lemma 1 (Scalar-to-tensor approximation)**.** *Let $\|\mathcal{S}^\star\|_\infty \leq B$. For any scalar function g and any degree-P polynomial q_P,*

$$\|g(\mathcal{S}^\star) - q_P(\mathcal{S}^\star)\|_F \leq \sqrt{\prod_{n=1}^{N} I_n} \sup_{|s|\leq B} |g(s) - q_P(s)|. \tag{20}$$

Proof. Every entry of $\mathcal{S}^\star$ lies in $[-B, B]$, so every entrywise approximation error is bounded by $\sup_{|s|\leq B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the claim. $\square$

Proof of Theorem 2. Let $g(s) = \sum_{k=0}^{\infty} a_k s^k$ be the Taylor expansion of g around zero, and choose $q_P(s) = \sum_{k=0}^{P} a_k s^k$. By Cauchy's estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \tag{21}$$

For $|s| \leq B < \rho$,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^{\infty} |a_k|\,|s|^k \tag{22}$$

$$\leq M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \tag{23}$$

Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^\star$, $\beta_k = a_k$ for $0 \leq k \leq P$, yields the stated unnormalized bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^{N} I_n$, gives the normalized form used in Theorem 2. $\square$

B.2 Rank Expansion

Proof of Proposition 1. Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1,\dots,i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1,\dots,r_N} \prod_{n=1}^{N} A^{(n)}_{i_n,r_n}. \tag{24}$$

Taking an element-wise p-th power gives

$$(\mathcal{S}_{i_1,\dots,i_N})^p = \prod_{q=1}^{p} \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}} \prod_{n=1}^{N} A^{(n)}_{i_n,r_n^{(q)}} \right) \tag{25}$$

$$= \sum_{\gamma_1\in[R_1]^p} \cdots \sum_{\gamma_N\in[R_N]^p} \left(\prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}} \right) \prod_{n=1}^{N} \left(\prod_{q=1}^{p} A^{(n)}_{i_n,r_n^{(q)}} \right), \tag{26}$$

where $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$. For each mode n, let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n, so $|\alpha_n| = p$. Then

$$\prod_{q=1}^{p} A^{(n)}_{i_n,r_n^{(q)}} = \prod_{r=1}^{R_n} \left(A^{(n)}_{i_n,r}\right)^{\alpha_{n,r}} = A^{(n)\langle p\rangle}_{i_n,\alpha_n}. \tag{27}$$

Group all ordered tuples $(\gamma_1, \dots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \dots, \alpha_N)$, and define the aggregated core

$$\mathcal{X}^{\langle p \rangle}_{\alpha_1,\dots,\alpha_N} = \sum_{\substack{\gamma_1,\dots,\gamma_N: \\ \mathrm{count}(\gamma_n)=\alpha_n \ \forall n}} \prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}}. \tag{28}$$

The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = [\![\mathcal{X}^{\langle p \rangle}; \mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle}]\!]. \tag{29}$$

The n-th Veronese factor has $\binom{R_n+p-1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

B.3 Separation

Lemma 2 (Full-rank Veronese evaluations)**.** *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D\times R}$ whose Veronese feature matrix $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{D\times D}$ has rank D. Consequently, for $I \geq D$, a generic matrix $\mathbf{A} \in \mathbb{R}^{I\times R}$ has $\mathrm{rank}(\mathbf{A}^{\langle p \rangle}) = D$.*

Proof. The homogeneous degree-p monomials in R variables are linearly independent polynomials. Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^\alpha)_{i,\alpha}$, $|\alpha| = p$, is nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the first claim.

For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is therefore a generic property, so $\mathbf{A}^{\langle p \rangle}$ has full column rank for generic $\mathbf{A}$. $\square$

Proof of Theorem 3. Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in \mathbb{R}^{I_n\times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^{R} a_r^{(1)} \otimes \cdots \otimes a_r^{(N)}. \tag{30}$$

Its mode-n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \cdots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \cdots \odot \mathbf{A}^{(1)} \right)^\top. \tag{31}$$

For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R, and the Khatri-Rao product on the right has rank R. Thus $\mathrm{rank}(\mathcal{S}_{(n)}) = R$ for every n, so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree-p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1,\dots,i_N} = \left(\sum_{r=1}^{R} \prod_{n=1}^{N} A^{(n)}_{i_n,r} \right)^p \tag{32}$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^{N} \prod_{r=1}^{R} \left(A^{(n)}_{i_n,r} \right)^{\alpha_r}, \tag{33}$$

where $c_\alpha = p!/(\alpha_1! \cdots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n\times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full column rank D for generic $\mathbf{A}^{(n)}$. The mode-n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \,\mathrm{diag}(c) \left(\mathbf{B}^{(N)} \odot \cdots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \cdots \odot \mathbf{B}^{(1)} \right)^\top. \tag{34}$$

The diagonal matrix $\mathrm{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices is full column rank, the right factor has rank D. Therefore $\mathrm{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every mode n, and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

B.4 Masked Generalization

Theorem 4 (Masked-entry uniform convergence)**.** *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set with*

$$d = \prod_{n=1}^{N} R_n + \sum_{n=1}^{N} I_n R_n + (P+1).$$

For $\theta \in \Theta$, define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \qquad M = \prod_{n=1}^{N} I_n,$$

and the masked empirical risk

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^{m} \ell_{\mathbf{i}_t}(\theta),$$

where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \le \ell_{\mathbf{i}}(\theta) \le B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G-Lipschitz in θ, uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ, then with probability at least $1-\delta$,

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

for every $0 < \varepsilon \le B_\Theta$.

Proof of Theorem 4. Fix $0 < \varepsilon \le B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε-net of Θ in Euclidean norm. Since Θ lies in a d-dimensional ball of radius B_Θ, it admits a net with

$$|\mathcal{N}_\varepsilon| \le \left(\frac{3B_\Theta}{\varepsilon} \right)^d. \tag{35}$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{\mathbf{i}_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr\left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \ge u \right) \le 2 \exp\left(-\frac{2mu^2}{B_\ell^2} \right). \tag{36}$$

Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1-\delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \le B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \tag{37}$$

Substituting the net-size bound yields the stated stochastic term.

Now take any $\theta \in \Theta$, and choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \le \varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \le G\varepsilon, \qquad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \le G\varepsilon. \tag{38}$$

Therefore

$$\left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \tag{39}$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \tag{40}$$

$$\le 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \tag{41}$$

Taking the supremum over $\theta \in \Theta$ completes the proof. □

Remark 1. *The Lipschitz and bounded-loss assumptions are standard for compact parametric classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may depend polynomially on these bounds and on P, but the covering dimension remains the number of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker expansion.*

C Additional Experiments

C.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 5 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^\star)$. Let $S^\star$ be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^\star = g(S^\star) - \frac{\langle g(S^\star), S^\star \rangle}{\|S^\star\|_F^2} S^\star, \tag{42}$$

and, when $\|R^\star\|_F > 0$, normalize it as

$$\widetilde{R}^\star = \frac{\|S^\star\|_F}{\|R^\star\|_F} R^\star. \tag{43}$$

The observed tensor is then

$$Y_\alpha = \sqrt{1-\alpha}\, S^\star + \sqrt{\alpha}\, \widetilde{R}^\star, \qquad 0 \le \alpha \le 1. \tag{44}$$

By construction, $\langle S^\star, \widetilde{R}^\star \rangle = 0$ and $\|\widetilde{R}^\star\|_F = \|S^\star\|_F$, so α is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are $g(s) = s^2$, $g(s) = s^2 + s^3$, $g(s) = \sin(\kappa s)$, and $g(s) = \tanh(\kappa s)$, with κ set from the input scale.

The main text reports the nonlinear alpha sweep and degree sweep. Figure 4 shows that degree continuation activates higher powers without destabilizing the fit. Table 6 complements those figures with a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

Table 5: Linear consistency on synthetic Tucker tensors. The $p > 1$ columns report effective higher-order NTD-PL contribution.

Method	bias $= 0$			bias $= 0.5$		
	RMSE↓	NMSE(dB)↓	$p > 1$	RMSE↓	NMSE(dB)↓	$p > 1$
Tucker	**0.0114**	**-38.88**	–	0.0647	-24.05	–
NTD-PL, $\beta_0 = 0$	0.0116	-38.72	4.19%	0.0522	-25.85	39.17%
NTD-PL, β_0 free	0.0117	-38.65	3.19%	**0.0118**	**-38.58**	2.98%

Table 6: Representative effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at the matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p=0$	$p=1$	$p=2$	$p=3$	$p=4$	$p=5$	$p=6$
s^2	0.5	43.3	**23.2**	5.8	8.2	9.1	9.8
s^2+s^3	0.1	24.1	9.8	**38.1**	7.3	11.5	9.1
$\sin(\kappa s)$	0.1	37.7	0.6	**35.4**	2.8	19.8	3.6
$\tanh(\kappa s)$	0.2	37.9	1.3	**30.9**	5.4	17.3	7.0

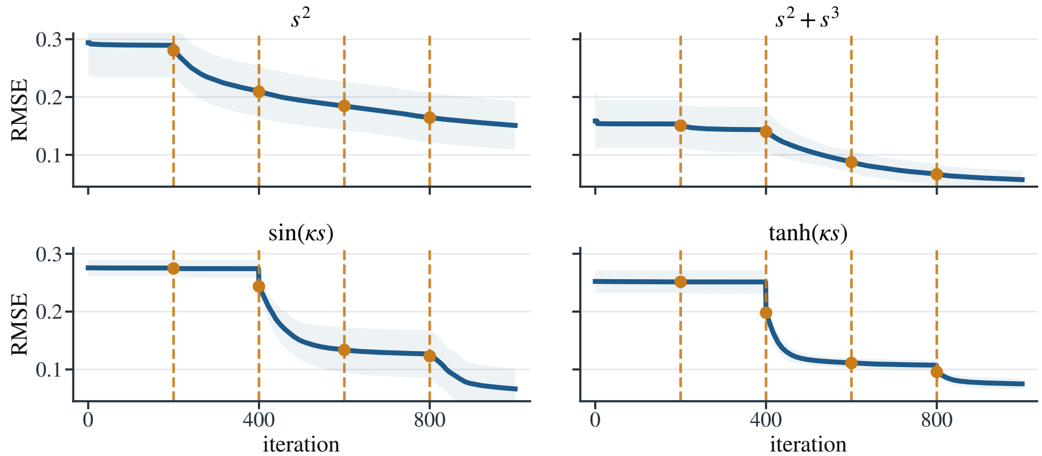

Figure 4: NTD-PL training curves for the controlled nonlinear mechanism checks at $\alpha = 0.25$ and $P = 5$. Dashed vertical lines mark degree-activation events.

C.2 CAVE Completion Significance and Robustness

The main text reports the full-resolution random-missing CAVE completion table at rank $(33, 33, 4)$. We do not repeat that table here; instead, Table 7 gives the paired sign and Wilcoxon tests for the same protocol.

Table 7: Significance tests for full-resolution CAVE completion at rank $(33, 33, 4)$. Metrics are computed on missing entries.

	RMSE				SAM			
ρ	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

C.3 Post-Hoc Calibration Implementation

The calibration baselines in Figure 5 are post-processing models applied to a fixed Tucker reconstruction. Let $\widehat{\mathcal{Y}}_{\mathrm{T}}$ denote the Tucker output and $\mathcal{Y}$ the target tensor. PolyCal fits a scalar degree-four polynomial

$$h_\gamma(x) = \sum_{p=0}^{4} \gamma_p x^p$$

by ridge regression on pairs $(\widehat{Y}_{\mathrm{T},\mathbf{i}}, Y_{\mathbf{i}})$. In full reconstruction, all entries are used for this fit. In completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted map is applied entrywise to $\widehat{\mathcal{Y}}_{\mathrm{T}}$; the Tucker core and factors are never updated.

MLPCal uses the same fixed-input protocol, but replaces the polynomial with a one-dimensional MLP. The input and target scalars are standardized, then a single-hidden-layer tanh network with a scalar skip connection is trained with Adam and ℓ_2 weight decay on the observed calibration pairs. In the CAVE runs we use 16 hidden units, learning rate 10^{-3}, weight decay 10^{-5}, batch size 8192, at most 200,000 calibration samples, and 1500 optimization steps. Thus both PolyCal and MLPCal test post-hoc scalar remapping of an already learned Tucker tensor, while NTD-PL jointly updates the latent Tucker geometry under the nonlinear link.

C.4 CAVE Beads Pseudo-RGB Visualization

Figure 6 visualizes the CAVE beads scene under the reconstruction and random-completion protocols. Each 31-band tensor is rendered as pseudo-RGB by selecting three representative spectral bands near

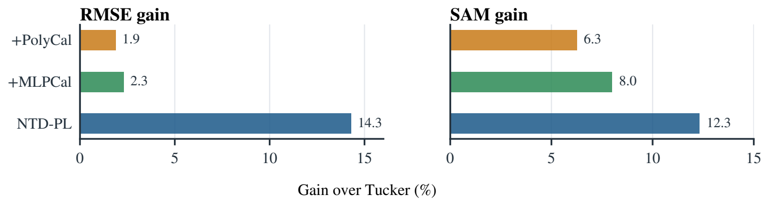

Figure 5: Post-hoc scalar calibration versus joint linked fitting on CAVE reconstruction at rank $(33, 33, 4)$. Gains are relative to Tucker.

75%, 50%, and 20% of the wavelength index range and applying a shared linear intensity scaling within each task. For completion, unobserved entries in the observation panel are shown in gray.

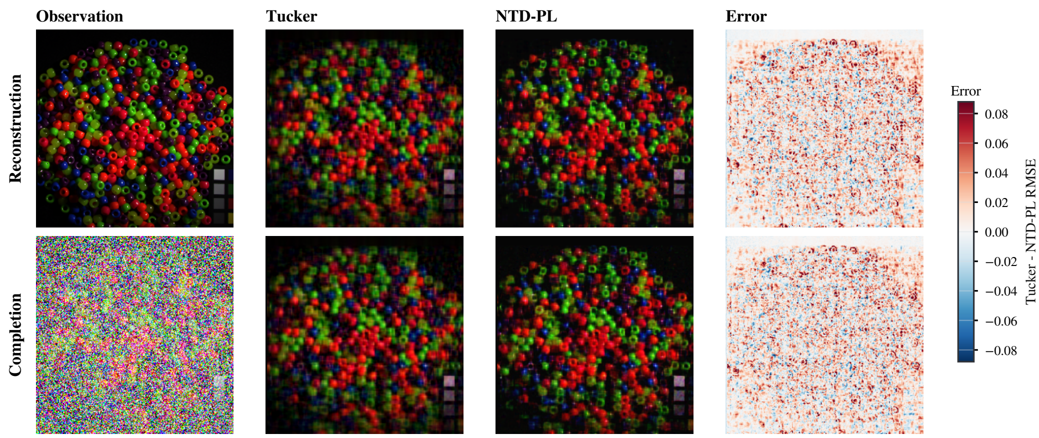

Figure 6: CAVE beads visual comparison. Rows correspond to reconstruction and random completion at missing rate 0.5. Error maps show Tucker minus NTD-PL per-pixel RMSE, so positive values indicate lower NTD-PL error.

C.5 Cross-Domain Real Tensor Checks

Table 8 reports matched-rank reconstruction on tensors outside the main hyperspectral setting: additional hyperspectral scenes, natural-image collections, and object-view tensors. We use NMSE(dB) as the main cross-dataset metric because it normalizes by the energy of each tensor and is therefore less sensitive to scale differences across domains. The results are consistent with the scoped claim in Section 6.4: gains are clearest on hyperspectral and object-view data, while natural-image collections provide more modest improvements.

Table 8: Additional matched-rank real tensor reconstruction. NMSE columns are in dB; lower is better. ΔNMSE is the Tucker-minus-NTD-PL dB gap, and NMSE gain is the relative reduction in linear NMSE; positive values favor NTD-PL.

Domain	Dataset	Shape	Rank	NMSE(dB)↓			NMSE gain
				Tucker	NTD-PL	Δ	
Hyperspectral	Jasper Ridge	$100\times100\times198$	(11,11,43)	-15.96	-16.58	**0.62**	**13.4%**
Hyperspectral	Samson	$95\times95\times156$	(10,10,35)	-19.37	-20.09	**0.72**	**15.3%**
Hyperspectral	Urban	$307\times307\times162$	(43,43,47)	-13.21	-13.53	**0.32**	**7.1%**
Natural images	CBSD68	$68\times96\times96\times3$	(20,32,32,3)	-11.27	-11.40	**0.13**	**3.0%**
Natural images	CIFAR-10	$1000\times32\times32\times3$	(96,20,20,3)	-17.04	-17.37	**0.34**	**7.5%**
Object views	COIL-100	$8\times36\times48\times48\times3$	(4,8,12,12,2)	-10.17	-10.71	**0.54**	**11.7%**
Object views	smallNORB	$25\times18\times64\times64\times2$	(18,12,24,24,2)	-25.97	-26.45	**0.48**	**10.5%**

NeurIPS Paper Checklist

This draft uses a working checklist. It should be replaced by the official NeurIPS checklist distributed with the final 2026 submission template.

1. **Claims.** The main claims are stated in the abstract, introduction, theory section, and experiment captions: NTD-PL is a parameter-efficient nonlinear observation model for Tucker tensors; its gains are not explained by modest rank increases or post-hoc scalar calibration.
2. **Limitations.** Section 7 discusses near-linear boundary cases, whole-axis missingness, shared-link assumptions, and the current domain concentration in hyperspectral data.
3. **Theory assumptions.** The approximation results assume a bounded latent tensor and polynomial or analytic scalar links. The separation results use generic full-rank feature constructions. The masked generalization bound assumes bounded parameters, bounded observations, and random observed entries.
4. **Reproducibility.** The experiment section reports datasets, ranks, mask types, missing rates, and baseline definitions. The codebase includes scripts for rank sweeps, MLPCal, structured missingness, and the cross-domain real-tensor reconstruction benchmark; the final version should add hardware details and exact command manifests.